

MagMat-Discovery

Physics-Informed Machine Learning and Screening for Magnetic Materials

System Architecture, Methodology & Model Validation

Technical Report & Model Guide

Author: Sachin Poudel

Project: MagMat-Discovery Platform

Web Platform: Interactive Streamlit Discovery Engine

Date: September 2026

Contents

Executive Summary	1
Glossary of Key Terms	2
1 Data and Partitioning	3
1.1 Data Sources	3
1.2 Data Cleaning and Grouped Splits	3
2 Methodology	4
2.1 Architecture and Provenance	4
2.2 Input Representations and Crystal Generation	4
2.3 Ensembles, Cascades, and Uncertainty	5
3 Model Validation	6
3.1 Benchmark Performance	6
3.2 Parity Analysis	6
3.3 Diagnostics and Screening	7
4 Interactive Platform	10
5 Summary and Outlook	10
A Physical Feature Formulations	12
B MagFormer Architecture Details	12

Executive Summary

Discovering permanent magnets with reduced rare-earth content is important for electric vehicles, wind turbines, and industrial machinery. Useful candidates must meet three criteria: Curie temperature $T_C > 400$ K, saturation magnetization $\mu_0 M_s > 1.0$ T, and coercivity $H_c > 800$ kA/m. However, machine learning for magnetic materials is challenging because intrinsic properties depend on atomic chemistry while extrinsic properties depend on grain microstructure. Furthermore, random data splits cause leakage between identical compositions, and unconstrained models can make unphysical predictions.

MagMat-Discovery addresses these challenges through two primary objectives:

1. **Leak-Free Multi-Property Modeling:** Simultaneously predict 8 intrinsic and extrinsic magnetic properties (phase, transition temperatures, and hysteresis behavior) using domain physics features, graph models, and 5-fold grouped cross-validation to prevent chemical formula leakage.
2. **Physics-Constrained Screening and Uncertainty:** Connect intrinsic predictions to extrinsic properties via micromagnetic relations, quantify prediction confidence using 90% conformal intervals, and provide an interactive web platform for candidate discovery.

By combining experimental data, physical features, the **MagFormer** graph model, ensembles, and conformal calibration, the platform accurately predicts magnetic properties and provides physically bounded rankings for magnet discovery.

Keywords: Magnetic Materials, Physics-Informed Machine Learning, Graph Transformers, Conformal Calibration, Coercivity, Permanent Magnets, Materials Screening.

Glossary of Key Terms

Table 1 summarizes the primary magnetic properties and machine learning terminology used throughout this report.

Table 1: Glossary of magnetic properties, physical units, and computational terms.

Term	Unit	Definition
T_C (Curie Temp.)	K	Temperature above which a ferromagnet loses spontaneous magnetization.
T_N (Néel Temp.)	K	Temperature above which an antiferromagnet becomes paramagnetic.
M_s (Saturation Mag.)	emu/g or T	Maximum magnetization in an applied field, reported in emu/g or $\mu_0 M_s$ (T).
H_c (Coercivity)	A/m or kA/m	Magnetic field required to reduce magnetization to zero after saturation.
K_1 (Anisotropy Constant)	J/m ³	Energy density required to rotate magnetization away from the preferred crystal axis.
$(BH)_{\max}$ (Energy Product)	kJ/m ³	Maximum stored magnetic energy density; main figure of merit for permanent magnets.
θ_p (Paramagnetic Curie Temp.)	K	High-temperature intercept from Curie-Weiss susceptibility fitting.
FM / AFM / NM	Discrete	Ferromagnetic, Antiferromagnetic, or Non-magnetic ground-state ordering.
Out-of-Fold (OOF)	–	Predictions on held-out validation data, avoiding data leakage.
Conformal Range	Physical	Uncertainty window covering the true value 90% of the time.
dex	log ₁₀	Decimal order of magnitude. An error of 0.65 dex means a factor of $10^{0.65} \approx 4.5$.

1 Data and Partitioning

1.1 Data Sources

The platform uses three primary data sources:

- **NEMAD Database:** 51,394 experimental records [1] across all 8 target properties.
- **Materials Project:** Crystal symmetries (space groups 1–230, crystal systems, and densities) [2]. DFT energies and moments are excluded to prevent target leakage.
- **Bilbao MAGNDATA:** Magnetic space groups and Shubnikov symmetry classifications.

1.2 Data Cleaning and Grouped Splits

Chemical formulas are simplified to lowest integer ratios with `pymatgen` (e.g., $\text{Fe}_6\text{Pt}_2 \rightarrow \text{Fe}_3\text{Pt}$). Properties spanning multiple orders of magnitude (H_c , K_1 , $(BH)_{\max}$) are log-transformed.

To evaluate generalization to unseen materials, models are tested with 5-fold cross-validation grouped by chemical formula (`GroupKFold`). All entries with the same formula or crystal phase stay in the same split, ensuring test folds contain only new compounds.

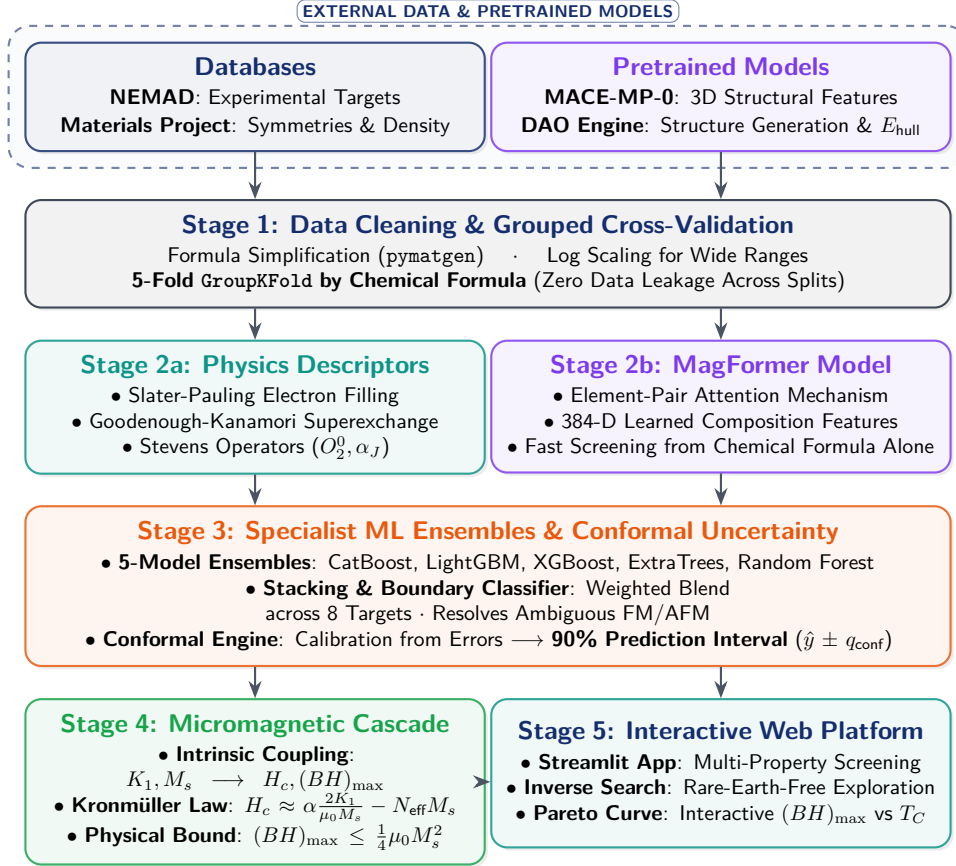


Figure 1: Workflow of MagMat-Discovery, showing external resources, in-house representations, 5-model ensembles, physical cascades, and candidate screening.

2 Methodology

2.1 Architecture and Provenance

Figure 1 shows the overall workflow of MagMat-Discovery, distinguishing external resources from in-house algorithms (Table 2). External databases and foundation models provide experimental records, crystal symmetries, and 3D crystal generation. All regressors, phase classifiers, micromagnetic cascades, and uncertainty calibrators were trained and evaluated in-house.

Table 2 lists the data sources, pretrained models, and in-house components used in the pipeline.

2.2 Input Representations and Crystal Generation

As outlined in Table 2, candidate materials are represented through domain physics features and graph neural embeddings:

- **Physics Descriptors**: Formulations covering composition, electronic structure, and single-ion anisotropy factors (detailed in Appendix A).
- **MagFormer Graph Embeddings**: Learned 384-dimensional representation vectors derived directly from chemical formulas via pair-biased self-attention (detailed in Appendix B).
- **Crystal Structure Synthesis**: When 3D atomic coordinates are required, structures are generated and evaluated for thermodynamic stability using the pretrained DAO framework [4].

Table 2: Data sources, models, and algorithmic components in MagMat-Discovery.

Component	Type	Source	Role in Pipeline
Materials Project [2]	External DB	Open DFT repository	Crystal symmetries and densities (energies/moments excluded).
NEMAD Database [1]	External DB	Experimental literature	Target measurements for all 8 magnetic properties.
MACE-MP-0 [3]	Pretrained	Equivariant GNN	3D structural features for compounds with known coordinates.
DAO Engine [4]	Pretrained	Diffusion model	Generates 3D unit cells (DAO-G) and checks stability E_{hull} (DAO-P).
MagFormer	In-House	Trained in this work	Maps chemical formulas to 384-D features without 3D coordinates.
Physics Descriptors	In-House	Domain physics	Electron counts (VEC), superexchange proxies, Stevens factors.
5-Model Ensembles	In-House	5-fold grouped training	Blends CatBoost, LightGBM, XGBoost, ExtraTrees, Random Forest.
Boundary Classifier	In-House	Ambiguous data	Resolves ambiguous FM vs. AFM phase predictions.
Micromagnetic Cascade	In-House	Micromagnetic theory [5]	Enforces Kronmüller coercivity and $(BH)_{\text{max}} \leq \frac{1}{4}\mu_0 M_s^2$.
Conformal Engine	In-House	Validation residuals [6]	Provides 90% prediction intervals for continuous properties.

2.3 Ensembles, Cascades, and Uncertainty

Target properties are predicted using 5-model ensembles combined through non-negative least squares weighting (Table 2), with ambiguous magnetic phases resolved by a specialized boundary classifier.

To ensure predictions obey physical laws, the pipeline enforces a micromagnetic cascade linking intrinsic parameters to extrinsic hysteresis behavior:

1. Intrinsic properties (K_1 and M_s) are predicted first.
2. These feed into coercivity (H_c) modeling to follow the Kronmüller relationship [5]:

$$H_c \approx \alpha_{\text{micro}} \frac{2K_1}{\mu_0 M_s} - N_{\text{eff}} M_s \quad (1)$$

where α_{micro} represents microstructural defect pinning and N_{eff} is the effective demagnetization factor.

3. Predicted energy products are bounded by the theoretical maximum: $(BH)_{\text{max}} \leq \frac{1}{4}\mu_0 M_s^2$.

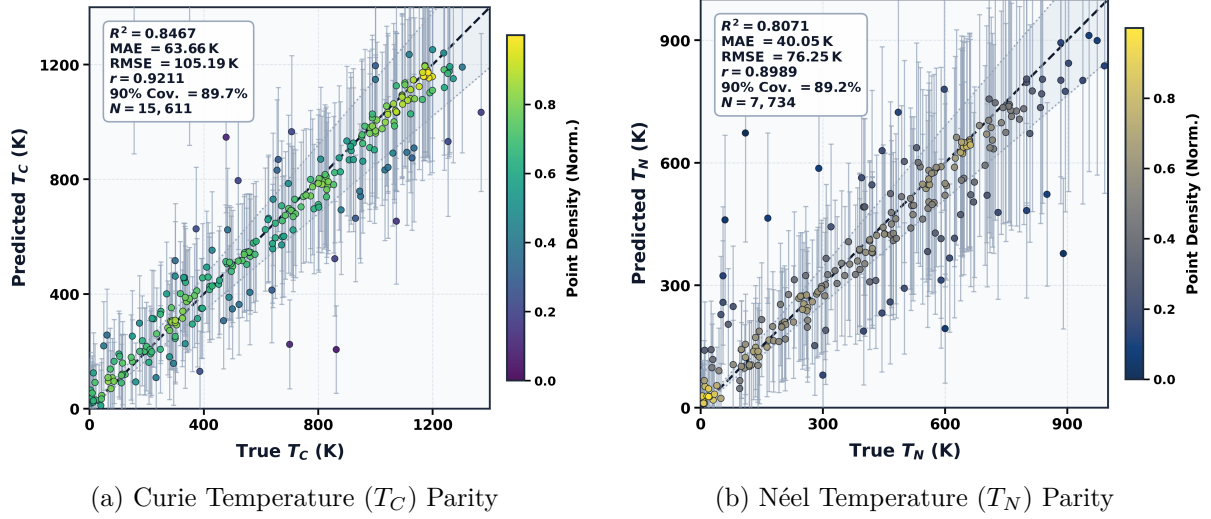
Every continuous property is accompanied by a distribution-free uncertainty estimate calibrated on held-out validation residuals [6]:

$$q_{\text{conf}} = \text{Quantile}_{0.90}(\{|y_i - \hat{y}_i|\}_{i=1}^n) \quad (2)$$

providing guaranteed 90% empirical coverage intervals $[\hat{y} - q_{\text{conf}}, \hat{y} + q_{\text{conf}}]$ for candidate screening.

Table 3: Comprehensive 5-fold GroupKFold cross-validation results across all 8 magnetic targets.

#	Target Property	Unit	Scale	Count n	Primary Metric	Mean Error	Practical Use
1	Ground-State Phase	Discrete	Class	35,045	91.9% Accuracy	Macro F_1 : 0.908	Strong (Reliable phase filtering)
2	Curie Temperature (T_C)	K	Linear	15,518	$R^2 = 0.816$	MAE = 69.8 K	Strong (Thermal threshold filter)
3	Néel Temperature (T_N)	K	Linear	7,734	$R^2 = 0.741$	MAE = 48.6 K	Strong (Accurate AFM boundary)
4	Anisotropy Constant (K_1)	J/m ³	Symlog	3,938	$R^2 = 0.461$	MAE = 0.95 dex	Moderate (Easy-axis candidate ranking)
5	Coercivity (H_c)	A/m	log ₁₀	6,980	$R^2 = 0.622$	MAE = 0.65 dex	Moderate (Hard vs soft screening)
6	Saturation Magnetization (M_s)	emu/g	Linear	7,619	$R^2 = 0.539$	MAE = 26.1 emu/g	Moderate (Moment estimation)
7	Energy Product ($(BH)_{\max}$)	kJ/m ³	log ₁₀	1,379	$R^2 = 0.507$	MAE = 0.27 dex	Moderate (Bounded density estimate)
8	Paramagnetic Curie Temp. (θ_p)	K	Linear	5,474	$R^2 = 0.607$	MAE = 64.7 K	Moderate (Exchange coupling sign)

Figure 2: Parity plots for Curie (T_C) and Néel (T_N) transition temperatures. Both models show strong accuracy across operating temperatures ($R^2 = 0.82$ and 0.74), with slight scatter below 100 K.

3 Model Validation

3.1 Benchmark Performance

Models are evaluated using 5-fold cross-validation grouped by chemical formula (Section 1.2). Performance is measured by the coefficient of determination (R^2) and mean absolute error (MAE). MAE is reported in physical units (K, emu/g) or decimal orders of magnitude (log₁₀ error) for wide-range properties (H_c , K_1 , $(BH)_{\max}$). Table 3 reports benchmark results across all 8 targets.

3.2 Parity Analysis

Performance reflects the difference between intrinsic properties (set by chemistry) and extrinsic properties (set by microstructure):

Transition Temperatures (T_C and T_N): Curie and Néel temperatures achieve high accuracy ($R^2 = 0.82$ and 0.74 , Figure 2) because ordering temperatures depend directly on chemical bonding and composition.

Magnetization and Coercivity (M_s and H_c): Saturation magnetization reaches $R^2 = 0.54$ (MAE = 26.1 emu/g, Figure 3). Coercivity spans four orders of magnitude (10^2 to 10^6 A/m). Because coercivity depends on grain boundaries and defects not captured by formula alone, predictions show wider scatter (MAE = 0.65 in log₁₀ scale). Nonetheless, the overall trend ($R^2 = 0.62$) reliably separates hard from soft magnets.

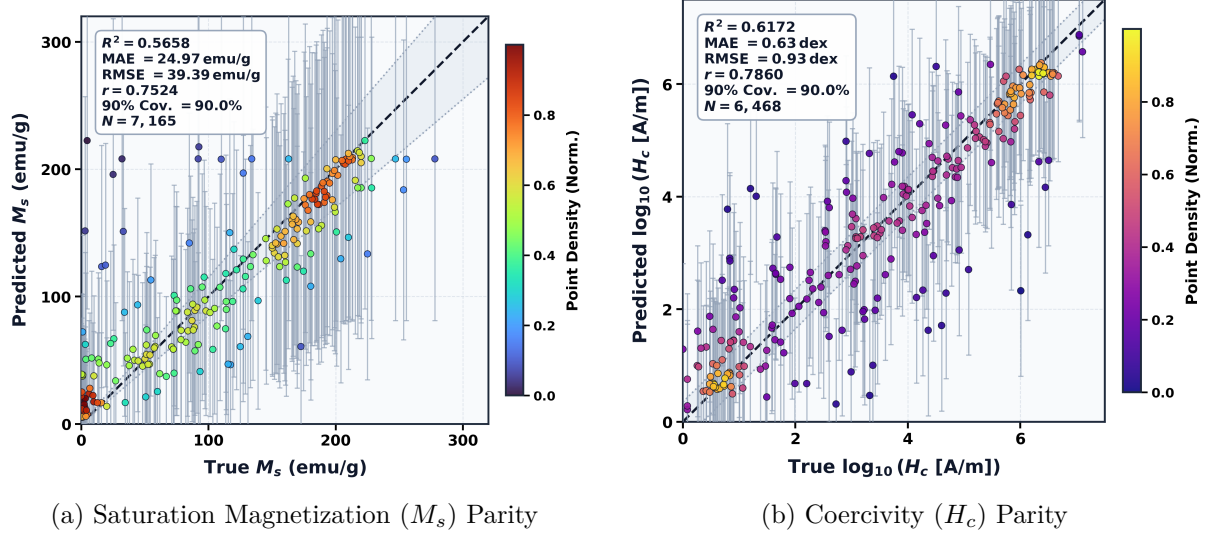


Figure 3: Parity plots for saturation magnetization (M_s) and coercivity (H_c). Magnetization shows moderate predictive utility ($R^2 = 0.54$). Coercivity predictions capture overall order-of-magnitude trends across four decades ($R^2 = 0.62$, MAE = 0.65 dex), serving as an effective pre-filter for hard magnetic behavior.

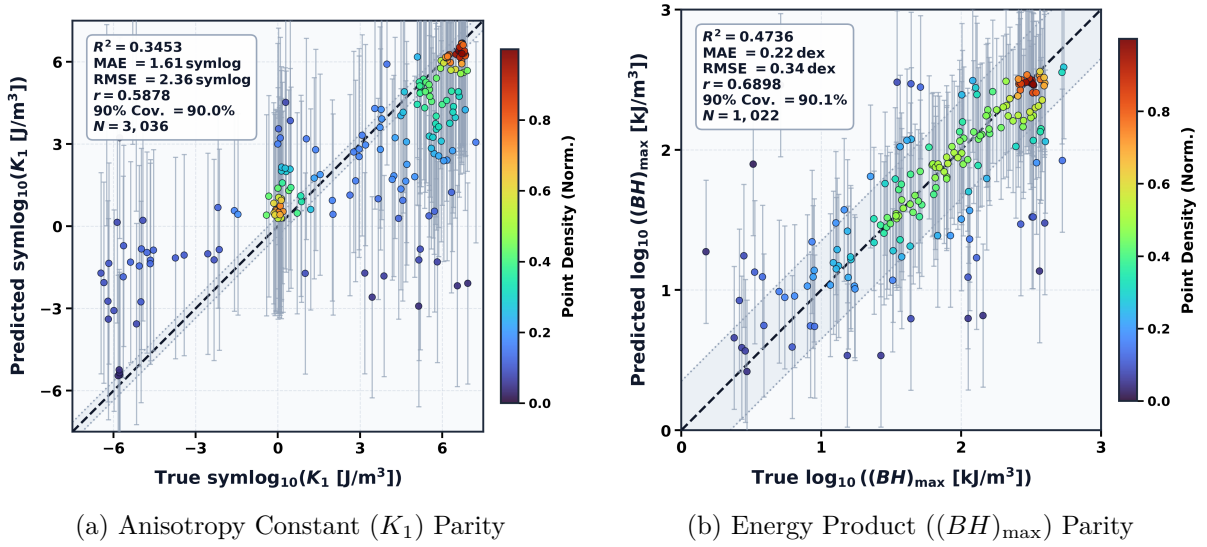


Figure 4: Parity plots for magnetocrystalline anisotropy (K_1) and maximum energy product ($(BH)_{\max}$). Cascade modeling ensures $(BH)_{\max}$ remains within thermodynamic limits. The moderate scores ($R^2 = 0.46$ and 0.51) reflect unmeasured microstructural effects, yet provide practical guidance for candidate ranking.

Anisotropy and Energy Product (K_1 and $(BH)_{\max}$): Anisotropy K_1 ($R^2 = 0.46$, Figure 4) identifies strong easy-axis candidates ($K_1 > 10^6$ J/m³). For energy product $(BH)_{\max}$ ($R^2 = 0.51$, MAE = 0.27 in $\log_{10}$ scale), the cascade enforces $(BH)_{\max} \leq \frac{1}{4}\mu_0 M_s^2$, preventing unphysical predictions.

3.3 Diagnostics and Screening

To assess representation quality, classification discrimination, statistical uncertainty, and practical candidate discovery:

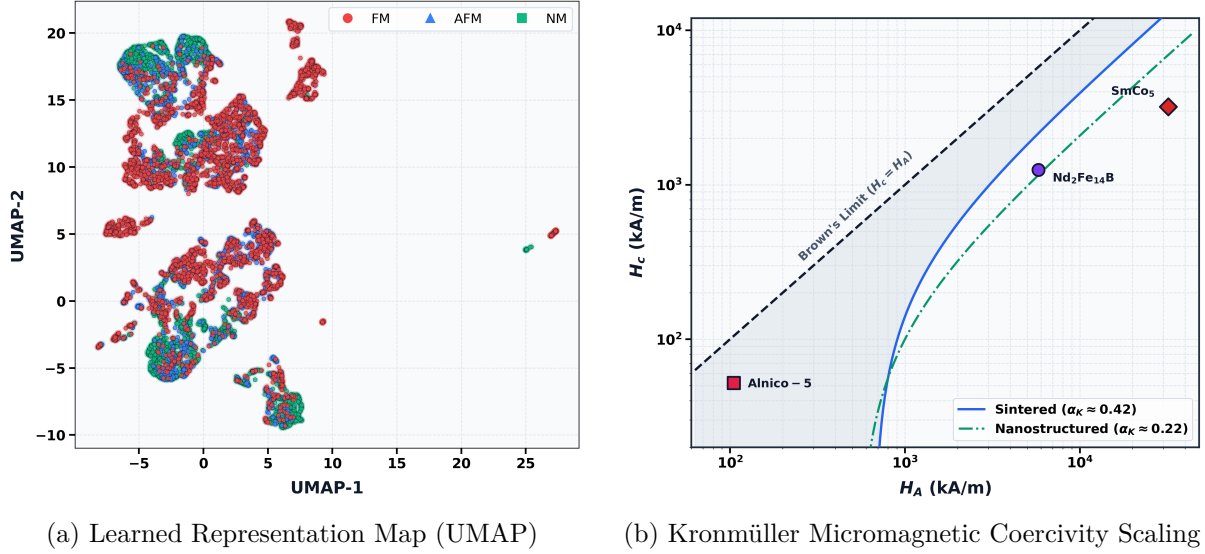


Figure 5: Latent representations and micromagnetic validation. (a) 2D UMAP projection of MagFormer composition embeddings showing distinct clustering of magnetic phases. (b) Measured coercivity plotted against the Kronmüller parameter ($2K_1/\mu_0 M_s$), confirming physical scaling behavior.

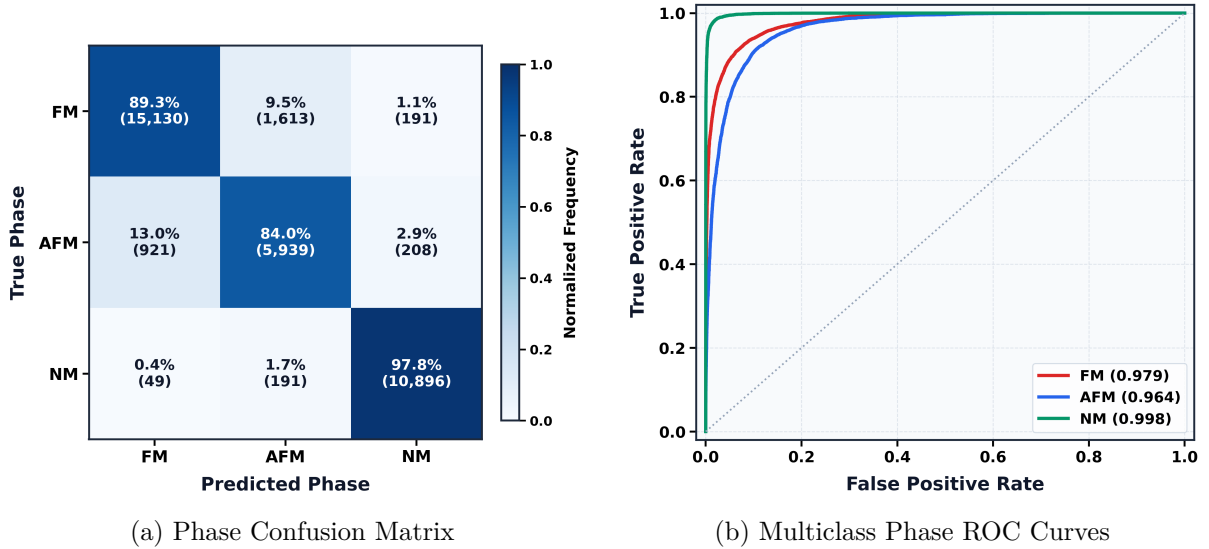
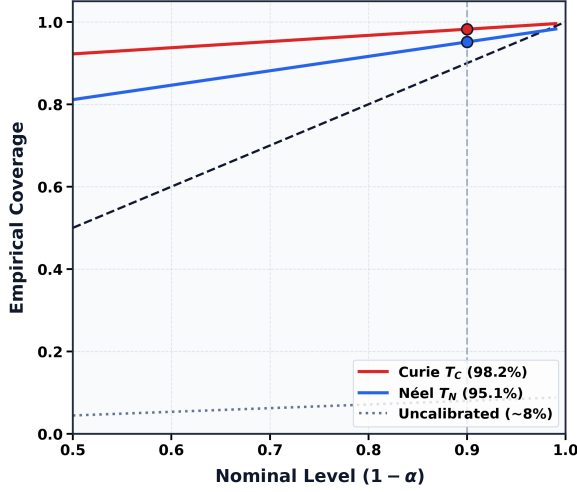


Figure 6: Ground-state magnetic phase classification diagnostics. (a) Confusion matrix showing balanced discrimination across phases. (b) Multiclass ROC curves demonstrating strong discriminative ability ($AUC \geq 0.96$) across Ferromagnetic, Antiferromagnetic, and Non-magnetic classes.

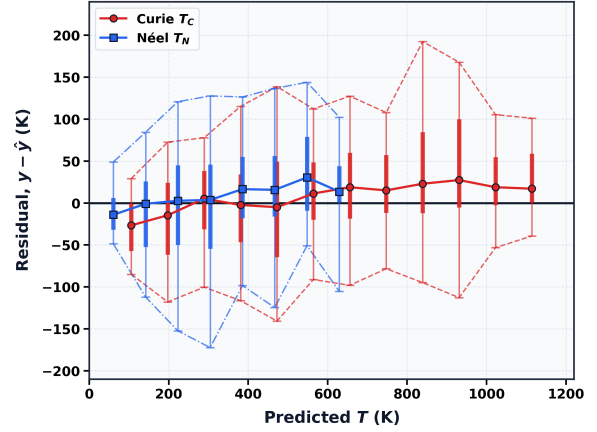
Learned Representations and Micromagnetic Scaling: Figure 5(a) maps MagFormer features via UMAP, showing distinct clusters for ferromagnetic, antiferromagnetic, and non-magnetic materials. Figure 5(b) confirms that coercivity scales with the Kronmüller parameter ($2K_1/\mu_0 M_s$).

Magnetic Phase Classification: Phase classification achieves 91.9% accuracy ($AUC \geq 0.96$, Figure 6), with the boundary classifier reaching $F_1 = 0.82$ on the smaller antiferromagnetic class.

Uncertainty Calibration: The conformal calibration curve (Figure 7(a)) closely tracks the ideal diagonal, confirming that 90% intervals achieve correct coverage. Residuals remain

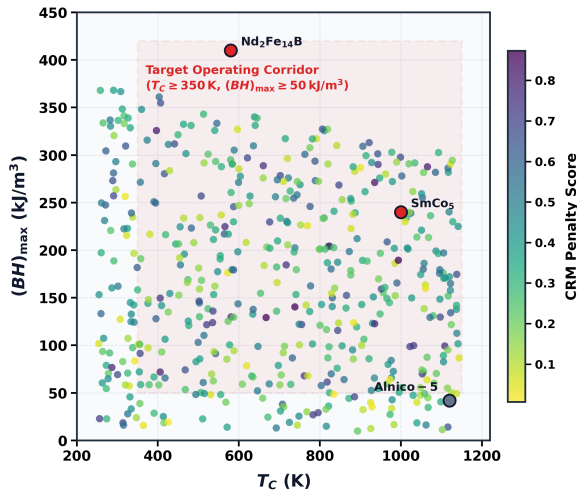
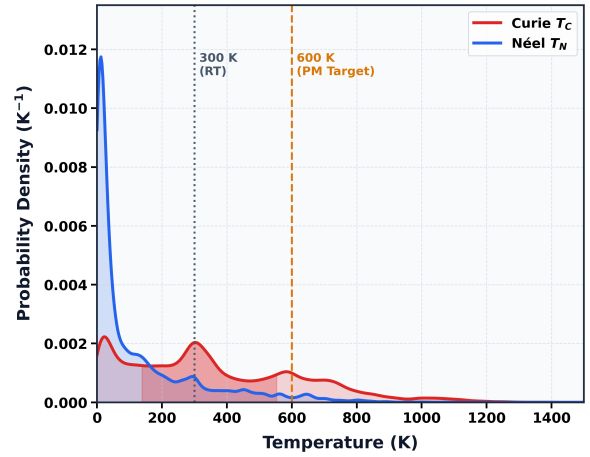


(a) Conformal Calibration Coverage Curve



(b) Residual Distributions across Temperature

Figure 7: Uncertainty calibration and residual diagnostics. (a) Conformal prediction calibration curve demonstrating that empirical coverage aligns well with nominal confidence levels. (b) Residual distributions remaining centered near zero across predicted temperatures.

(a) Pareto Frontier ($(BH)_{\max}$ vs T_C)

(b) Transition Temperatures by Crystal System

Figure 8: Candidate screening and symmetry profiles. (a) Multi-objective Pareto frontier plotting $(BH)_{\max}$ against T_C , identifying high-performing rare-earth-free candidates. (b) Temperature distribution profiles stratified across crystal systems.

centered near zero across operating temperatures (Figure 7(b)).

Candidate Screening: The Pareto frontier (Figure 8(a)) identifies candidates balancing $(BH)_{\max}$ and T_C . Crystal system distributions (Figure 8(b)) confirm that tetragonal and hexagonal systems provide the highest share of high- T_C , high-anisotropy materials.

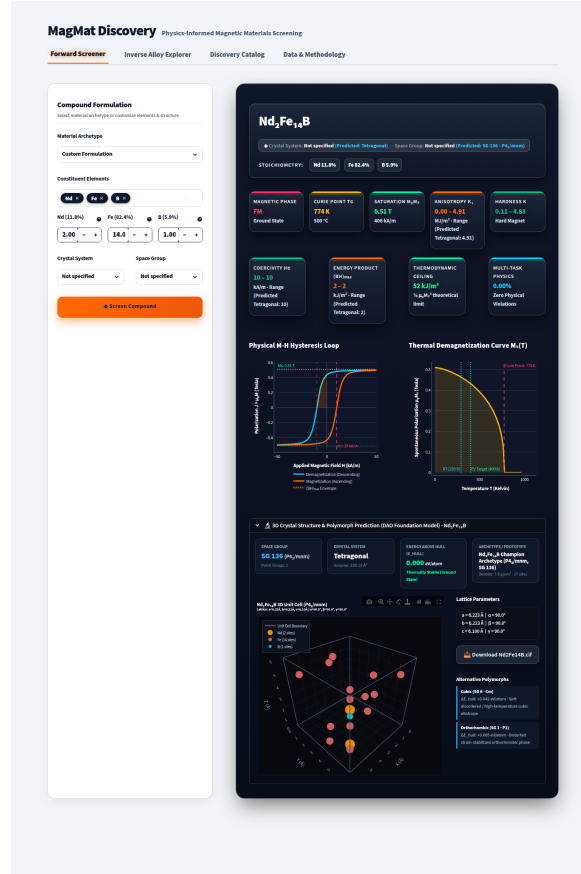


Figure 9: MagMat-Discovery interactive web platform interface. The forward screening dashboard displays predicted magnetic properties, multi-task physical consistency checks, simulated M - H hysteresis loop, thermal demagnetization curve $M_s(T)$, and interactive 3D crystal structure for $\text{Nd}_2\text{Fe}_{14}\text{B}$.

4 Interactive Platform

The MagMat-Discovery web application provides an interactive interface for candidate evaluation and screening (Figure 9), organized into four primary modules:

- **Forward Screener:** Users enter a formula or select reference magnets ($\text{Nd}_2\text{Fe}_{14}\text{B}$, SmCo_5 , Alnico, Fe_{16}N_2) to view property predictions with 90% intervals, simulated M - H loops, $M_s(T)$ curves, and 3D crystal structures.
- **Inverse Explorer:** Users set targets (e.g., $T_C \geq 500$ K, $(BH)_{\max} \geq 120$ kJ/m³) and filter by composition, such as excluding critical rare earths (Nd, Dy, Sm).
- **Discovery Catalog:** Browsing and export of material records, with filtering along the Pareto efficiency frontier.
- **Reference Tab:** Dataset statistics, space group distributions, and validation metrics displayed in the app.

5 Summary and Outlook

MagMat-Discovery combines physical principles, graph neural networks, and micromagnetic rules to screen permanent magnets:

- **Accurate Screening:** Magnetic phase classification (91.9% accuracy) and ordering temperatures ($R^2 = 0.82$ for T_C) serve as reliable initial filters.
- **Physical Ranking:** Extrinsic properties ($H_c, K_1, (BH)_{\max}$) achieve $R^2 = 0.46$ – 0.62 , providing a grounded pre-filter to prioritize candidates for synthesis.

Future work will incorporate grain-boundary phase models, domain wall pinning, and integration with automated synthesis platforms.

References

- [1] NEMAD Consortium, *A non-equilibrium magnetic materials database and machine learning framework for magnetic materials discovery*, Nature Communications, **16**, 2025, doi:10.1038/s41467-024-55500-7.
- [2] A. Jain et al., *Commentary: The Materials Project: A materials genome approach to accelerating materials innovation*, APL Materials, **1**(1), 011002, 2013.
- [3] I. Batatia, D. P. Kovacs, G. Simm, C. Ortner, and G. Csanyi, *MACE: Higher order equivariant and message passing neural networks for materials*, Advances in Neural Information Processing Systems (NeurIPS), **35**, 11423–11436, 2022.
- [4] Z. Huang et al., *Siamese Foundation Models for Crystal Structure Prediction*, Nature Communications, **17**, 2026, doi:10.1038/s41467-026-72362-3.
- [5] H. Kronmüller and M. Fähnle, *Micromagnetism and the Microstructure of High-Technology Materials*, Cambridge University Press, 2003.
- [6] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*, Springer, 2005.
- [7] J. M. D. Coey, *Magnetism and Magnetic Materials*, Cambridge University Press, 2010.

A Physical Feature Formulations

This appendix provides concise mathematical formulations for the domain descriptors listed in Table 2.

Goodenough-Kanamori Superexchange: Cation-anion-cation exchange coupling is parameterized as:

$$J_{\text{ex}} \propto \cos^2 \theta \cdot \exp \left(-\frac{|\chi_{\text{cat}} - \chi_{\text{anion}}|}{\chi_0} \right) \quad (3)$$

where θ is the bond angle, χ_{cat} and χ_{anion} are Pauling electronegativities, and $\chi_0 \approx 1.5$. Antiferromagnetic superexchange dominates near 180° , while ferromagnetic direct exchange is favored near 90° .

Dzyaloshinskii-Moriya Interaction (DMI) Proxy: Antisymmetric exchange in non-centrosymmetric crystals with heavy elements is modeled by:

$$D_{\text{proxy}} = \mathbb{I}(\text{non-centrosymmetric}) \sum_i x_i \left(\frac{Z_i}{100} \right)^4 \cdot \mathbb{I}(Z_i \geq 40) \quad (4)$$

where $\mathbb{I}(\cdot)$ is the indicator function, x_i is atomic fraction, and Z_i^4 reflects relativistic spin-orbit coupling scaling ($\lambda_{\text{SOC}} \propto Z^4$).

Crystal Field and Stevens Operator Equivalent: Uniaxial anisotropy (K_1) arising from single-ion crystal fields is parameterized by:

$$\mathcal{H}_{\text{CF}} = B_2^0 O_2^0 + B_4^0 O_4^0, \quad O_2^0 = 3J_z^2 - J(J+1) \quad (5)$$

where B_n^m are crystal-field parameters and J is total angular momentum. A negative Stevens factor ($\alpha_J < 0$) favors easy-axis anisotropy ($K_1 > 0$), while $\alpha_J > 0$ favors an easy plane.

B MagFormer Architecture Details

This appendix outlines the network equations for the MagFormer composition graph transformer.

Node Initialization: Each element i in a chemical formula is initialized via:

$$\mathbf{h}_i^{(0)} = \mathbf{W}_{\text{node}} [\mathbf{e}_{Z_i} \parallel x_i \parallel \mathbf{p}_i] \quad (6)$$

where $\mathbf{e}_{Z_i} \in \mathbb{R}^{64}$ is an element embedding, x_i is the atomic fraction, and $\mathbf{p}_i$ contains normalized elemental features (electronegativity, covalent radius, valence, Stoner parameter).

Pair-Bias Self-Attention: Inter-element relations are modeled via multi-head self-attention with pairwise relational bias:

$$\mathbf{A}_{ij} = \text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_j^T}{\sqrt{d_k}} + \mathbf{B}_{ij} \right) \mathbf{V}_j \quad (7)$$

where $\mathbf{Q}_i, \mathbf{K}_j, \mathbf{V}_j$ are linear projections of $\mathbf{h}^{(l)}$, and the relational bias $\mathbf{B}_{ij} = \text{MLP}(|\mathbf{p}_i - \mathbf{p}_j|)$ captures pairwise elemental differences.

Composition Pooling: A fixed 384-dimensional representation is pooled across the N constituent elements:

$$\mathbf{z}_{\text{comp}} = \left[\sum_{i=1}^N x_i \mathbf{h}_i^{(L)} \parallel \max_{i=1 \dots N} \mathbf{h}_i^{(L)} \parallel \frac{1}{N} \sum_{i=1}^N \mathbf{h}_i^{(L)} \right] \in \mathbb{R}^{384} \quad (8)$$

combining fraction-weighted, max, and mean pooling without task-specific target leakage.